

MONKES
(MONOENERGETIC KINETIC EQUATION SOLVER)
MANUAL
(DRAFT)

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August 18, 2025

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1 Installation and required libraries

In this section it is sketched how to install MONKES.

1.1 Fortran compiler

MONKES is written in Fortran language and thus requires having an installed Fortran compiler. This compiler must be compatible with LAPACK and NetCDF libraries. One possible compiler is the Intel Fortran compiler.

1.2 NetCDF library

In order to read BOOZER_XFORM output files, MONKES needs to use the NetCDF library. An example of the minimal required libraries for running MONKES in a UNIX machine is displayed below. Modules 1) and 2) are the NetCDF library compatible with the Fortran compiler. Module 3) is the Fortran compiler version, in this case intel17/17.0.8.

Currently Loaded Modules:

- 1) netcdf-fortran-4.4.4-intel-17.0.8-4me7upi
- 2) netcdf-4.6.1-intel-17.0.8-i5cn5xw
- 3) intel17/17.0.8

1.3 LAPACK library

For using LAPACK library there are two options. One can use the static or the dynamic version of the library. In listing 1, the dynamic case is illustrated.

```
# *** Compiler selection
f90comp=gfortran
NETCDF_DIR=/usr/include/

FFLAGS=-I$(NETCDF_DIR) -mmodel=large -O2 -freal-4-real-8 -ffree-line-length-512
LFLAGS=-llapack -lnetcdf -lnetcdf -lblas
```

Listing 1: Makefile

The variable LFLAGS includes the linking of LAPACK libraries. Specifically, those flags which include the variable \$MKL_HOME (variable containing the location of the dynamic library) set the instructions for where to find the library. The flag -mkl=parallel allows for LAPACK multithreading functionalities. In listing 2, it is shown how the monkes executable is generated.

```
main_monkes.x: $(objects)
    $(f90comp) -o main_monkes.x $(objects) ${LFLAGS} ${FFLAGS}
```

Listing 2: Makefile

2 How to use MONKES

2.1 Magnetic configuration input

MONKES can read the magnetic configuration of a particular flux surface from different types of files.

2.1.1 From BOOZER_XFORM output file

MONKES can read the magnetic configuration of an output (binary) file from BOOZER_XFORM. The file **must** be named boozmn.nc. It is advisable to make a soft link instead of renaming the original BOOZER_XFORM output file.

In an output file from BOOZER_XFORM there is information about the equilibrium of many flux surfaces. If not specified, MONKES will perform its calculations at the flux surface labelled by $s = \psi/\psi_{\text{lcfs}} = 0.25$. A different flux surface can be specified in the input file monkes_input.surface. An example on how to fill the input file monkes_input.surface is given in listing 3.

```
&surface
s=0.2
/
```

Listing 3: Example of `monkes_input.surface` for specifying the surface $s = \psi/\psi_{\text{lcs}} = 0.2$

2.1.2 From DKES input file

For running MONKES using a DKES input file, it **must** be renamed as `ddkes2.data`.

2.2 Monoenergetic database input

In addition to the magnetic configuration (see section 2.1), in order to run MONKES for calculating the monoenergetic coefficients $\hat{D}_{ij}$, it is necessary to specify 5 additional parameters. The required parameters are listed in table 1.

These parameters have to be specified in the input file `monkes_input.parameters`. For each combination of $(N_\theta, N_\zeta, N_\xi, \hat{\nu}, N_\xi)$ MONKES will compute the monoenergetic coefficients $\hat{D}_{ij}$. An example on how to fill this input file is given in listing 4. Note that in the example shown in listing 4 there is a single value for `N_theta`, `N_zeta` and `N_xi` while there are 10 values `nu` and 2 values of `E_r`. This means that MONKES will perform a scan in $\hat{\nu}$ and $\hat{E}_r$ for fixed resolution $(N_\theta, N_\zeta, N_\xi) = (30, 60, 160)$. Similarly, one could be interested in doing scans in the different resolutions to check how the monoenergetic coefficients $\hat{D}_{ij}$ converge. This scan is also possible by introducing several values of, say N_ζ , separated by commas.

```
&parameters
N_theta = 30
N_zeta = 60
N_xi = 160
nu =
1.000000e-05,
3.000000e-05,
1.000000e-04,
3.000000e-04,
1.000000e-03,
3.000000e-03,
1.000000e-02,
3.000000e-02,
1.000000e-01,
3.000000e-01
E_r = 0.000000e+00, 3e-4
/
```

Listing 4: Example of `monkes_input.parameters` for calculating a monoenergetic database

2.3 Monoenergetic database output

The monoenergetic coefficients $\hat{D}_{ij}$ are written in an output file called `monkes_Monoenergetic_Database.dat`. Instead of writing directly $\hat{D}_{ij}$ which are in meters, in the output file `monkes_Monoenergetic_Database.dat`, the quantities written are $K_{ij}\hat{D}_{ij}$. The monoenergetic coefficients are defined in section 2 of reference [1]. The quantities K_{ij} are defined in Appendix E of [1].

Parameter	Symbol in MONKES paper [1]	Name for MONKES	Units
Poloidal resolution	N_θ	<code>N_theta</code>	N/A
Toroidal resolution	N_ζ	<code>N_zeta</code>	N/A
No. Legendre modes	N_ξ	<code>N_xi</code>	N/A
Collisionality	$\hat{\nu}$	<code>nu</code>	1/m
Radial electric field	$\hat{E}_r$	<code>E_r</code>	V · m/s

Table 1: Parameters to be specified in the file `monkes_input.parameters` for calculating the monoenergetic coefficients.

nu/v [m ⁻¹]		E_r/v [V s /m ²]		N_theta		N_zeta		N_xi	
D_11		D_31		D_13		D_33		D_33_Spitzer	
time [s]	CPU time [s]								Wall-clock
0.1000000000000000E-04	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03		
0.8083370894646827E-01	0.3558097167954141E+00			-0.3557974181173633E+00	0.3781812078503804E+05	0.6628557164949892E+05			
0.4385000000000000E+02	0.5232387820000000E+03								
0.3000000000000000E-04	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.2810661936903304E-01	0.3256727047296022E+00			-0.3256699118937129E+00	0.1291631323161333E+05	0.2209519054983298E+05			
0.4500700000000000E+02	0.5381165510000001E+03								
0.1000000000000000E-03	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.9699155660889601E-02	0.2329025257396027E+00			-0.2329024600913502E+00	0.4016302448315862E+04	0.6628557164949895E+04			
0.4472800000000000E+02	0.5347238680000000E+03								
0.3000000000000000E-03	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.4279607172778349E-02	0.8845086936896115E-01			-0.8845086932979547E-01	0.1408020116218274E+04	0.2209519054983297E+04			
0.4602000000000000E+02	0.5501236660000002E+03								
0.1000000000000000E-02	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.2447678833565718E-02	-0.5447447639450777E-02			0.5447447639461694E-02	0.4599905179277028E+03	0.6628557164949892E+03			
0.4708800000000000E+02	0.5628036520000001E+03								
0.3000000000000000E-02	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.2456221430843609E-02	-0.3895792255643186E-01			0.3895792255645109E-01	0.1702769606149794E+03	0.2209519054983300E+03			
0.4869800000000000E+02	0.5820463040000000E+03								
0.1000000000000000E-01	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.3404582820706755E-02	-0.3962755023427194E-01			0.3962755023427266E-01	0.5800868190595923E+02	0.6628557164949895E+02			
0.4675700000000000E+02	0.5583137999999999E+03								
0.3000000000000000E-01	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.4750847233951999E-02	-0.2123155106854335E-01			0.2123155106854411E-01	0.2099408256854233E+02	0.2209519054983298E+02			
0.4713000000000000E+02	0.5633311179999996E+03								
0.1000000000000000E+00	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.8203825856789445E-02	-0.6314388928966534E-02			0.6314388928967790E-02	0.6540004603642041E+01	0.6628557164949894E+01			
0.4763800000000000E+02	0.5692957600000000E+03								
0.3000000000000000E+00	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.1908565050702955E-01	-0.1079732003337501E-02			0.1079732003338740E-02	0.2204175406018691E+01	0.2209519054983298E+01			
0.4679100000000000E+02	0.5591580839999997E+03								
0.1000000000000000E-04	0.3000000000000000E-03			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.1400431461308468E-01	0.2321125616633982E+00			-0.2321089946168132E+00	0.3785637549091421E+05	0.6628557164949892E+05			
0.4883100000000000E+02	0.5835083770000001E+03								
0.3000000000000000E-04	0.3000000000000000E-03			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.1560611355971947E-01	0.2809610046367516E+00			-0.2809607056510192E+00	0.1293144634657525E+05	0.2209519054983298E+05			
0.4701800000000000E+02	0.5618436950000005E+03								
0.1000000000000000E-03	0.3000000000000000E-03			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03			
0.8697565729005768E-02	0.2156076558873087E+00			-0.215					

Listing 5: Example of monkes_Monoenergetic_Database.dat for calculating a monoenergetic database

References

- [1] F.J. Escoto, J.L. Velasco, I. Calvo, M. Landreman, and F.I. Parra. Monkes: a fast neoclassical code for the evaluation of monoenergetic transport coefficients in stellarator plasmas. *Nuclear Fusion*, 64(7):076030, 2024. [doi:10.1088/1741-4326/ad3fc9](https://doi.org/10.1088/1741-4326/ad3fc9).